

CE 524P: Computational Hydroclimatology Lab

Course Code	CE 524P
Course Name	Computational Hydroclimatology Lab
L-T-P-C	0-0-2-1
Prerequisites	None
Intended for	PG and Ph.D. students
Approval	56th BoA (Jul 2024)

Laboratory Modules

1. **Acquiring the hydroclimatic data:** Acquisition and preprocessing of hydroclimatic data from various sources (2 Hours)
2. **Analysis of Climate Change Trends:** Analyze long-term temperature and precipitation data to identify and quantify climate change trends. (2 hours)
3. **Handling Missing Hydroclimatic Data:** Apply various techniques to handle missing data in hydroclimatic time series and evaluate their effectiveness. (2 hours)
4. **Simple Energy Balance Climate Model:** Implement a zero-dimensional energy balance climate model to simulate global temperature changes under different forcing scenarios. (2 hours)
5. **Hydrologic Modelling with Lumped Models:** Set up and calibrate multiple lumped models to characterize the hydrologic behavior of the watershed.(2 Hours)
6. **Hydrologic Modeling with SWAT:** Set up and run a SWAT model for a small watershed to simulate hydrological processes and assess impacts of land use changes. (2 hours)
7. **Downscaling and Bias Correction of Climate Data:** Apply statistical downscaling and bias correction techniques to GCM outputs for use in local impact studies. (2 hours)
8. **Analysis of Climate Indices:** Calculate and analyze various climate indices to understand large-scale climate patterns and their local impacts. (2 hours)
9. **Drought Analysis using Standardized Precipitation Index (SPI):** Compute SPI at multiple timescales to analyze historical drought patterns and assess future drought risk. (2 hours)
10. **Flood Frequency Analysis:** Perform flood frequency analysis using annual maximum streamflow data and assess non-stationarity due to climate change. (2 hours)
11. **Climate Change Impact on Crop Yield:** Assess the impact of climate change on crop yield using a simple crop model and climate projections. (2 hours)
12. **Glacier Mass Balance Modeling:** Implement a simple glacier mass balance model to assess climate change impacts on glacier evolution. (2 hours)

Text Books

1. None. Experiment notebooks will be provided during the lab

References

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